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SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

COURSE PROJECT REPORT

Course: Trustworthy AI

SPURIOUS CORRELATIONS AND INVARIANT LEARNING

ERM vs IRM vs REx on Colored-MNIST

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ABSTRACT

Machine learning models can perform well on in-distribution data but fail under distribution shift when they rely on spurious correlations. This report studies this problem using Colored-MNIST, where digit shape is the stable signal and color is spuriously correlated with the label during training but largely reversed at test time. We compare ERM, IRMv1, and V-REx under the same training and evaluation pipeline using multiple random seeds and train-domain validation as the primary model-selection protocol. Additional experiments examine the effects of penalty strength, backbone architecture, number of environments, additional spurious features, sample size, and dataset choice. The results show that ERM achieves high training accuracy but performs poorly on the OOD test set. IRMv1 improves average OOD accuracy, although its results vary substantially across seeds and checkpoint-selection rules. V-REx has a similar mean OOD accuracy to IRMv1, with less variation across seeds in the main comparison. The ablation experiments show that architecture and additional spurious features can substantially change the results. However, reducing the number of environments or the sample size does not consistently reduce average OOD accuracy. Calibration, attribution maps, recoloring tests, and linear probes provide further information about model behavior. Color remains decodable from both ERM and IRMv1 representations, while the attribution maps alone cannot establish whether either model relies more on shape. Overall, the results highlight the importance of model selection, repeated runs, and diagnostic analysis when evaluating OOD generalization under spurious correlations.

Keywords: spurious correlation, invariant learning, IRM, V-REx, OOD generalization, model selection, trustworthy AI.

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1. Introduction

Some machine learning models achieve high accuracy on in-distribution data; however, they may fail under distribution shift, resulting in poor performance in practical applications. One reason for this problem is that some features are correlated with the target label in the training data, but this correlation does not hold in real-world settings. This is called a *spurious correlation*. A common example of this phenomenon is the cow-camel classification problem, where cows are predominantly observed on grassy backgrounds while camels are associated with sandy backgrounds in the training data. As a result, a classifier may learn to rely on the background rather than the characteristics of the animal itself, leading to poor performance when this correlation changes at test time [1].

To study this problem, we set up a controlled environment using **Colored-MNIST** [2], where digit shape serves as the stable signal, while color is spuriously correlated with the label during training and anti-correlated at test time. This setting provides a simple way to observe how models behave when a spurious correlation is present and how different training objectives respond to changes in the data distribution. Based on this setup, the report focuses on the following research questions:

- **RQ1 (reproduce):** Does ERM rely on color and experience a drop in OOD performance, and how do invariant learning objectives behave under the same setting?
- **RQ2 (break):** Under what conditions does invariant learning become less effective, considering penalty strength, backbone capacity, number of environments, a second spurious feature, sample size, and dataset choice?
- **RQ3 (diagnose):** Which diagnostics can help indicate whether a model relies less on color, including calibration, attribution methods, and representation probes?

The report includes three main parts:

1. We compare ERM, IRMv1, and V-REx across three random seeds. Held-out train-domain validation is the primary checkpoint-selection rule. We also report OOD-oracle selection and the final checkpoint to examine how model selection affects the results [3].
2. We conduct 18 ablation settings covering the IRM-λ sweep, REx-β sweep, MLP and CNN backbones, single-environment training, a second spurious patch, small-n settings, and transfer to Colored-FashionMNIST
3. We examine model behavior using reliability analysis, Integrated Gradients, saliency maps, recoloring sensitivity, and linear probes, with particular attention to the Reliability and Explainability aspects of trustworthy AI

2. Related work

2.1. Invariant Risk Minimization (IRM)

Arjovsky et al. [2] introduce Invariant Risk Minimization (IRM), which aims to learn a representation Φ such that the same classifier w performs well across all training environments. In practice, the commonly used IRMv1 objective adds a gradient-based penalty to encourage this invariance (see §3). Later studies show that IRMv1 can be sensitive to the random seed and the penalty weight λ. Several extensions have therefore been proposed, including game-based optimization [4], gradient-variance regularization such as Fishr [5], and sparse variants of IRM [6].

2.2. Risk Extrapolation (REx) and Group DRO

V-REx [7] encourages similar risks across training environments by penalizing the variance of environment-wise risks. This provides a simpler alternative to the IRMv1 penalty. Group DRO [8] takes a different approach by minimizing the loss of the worst-performing environment. The implementation also includes a simplified Group DRO objective, but it is not included in the experimental comparisons reported here.

2.3. Limitations of invariant learning

Previous work also studies cases where invariant learning may not work as expected. Rosenfeld et al. [9] show that IRM requires enough training environments relative to the number of spurious features, and that learning from a single environment is not sufficient in their setting. Zhou et al. [6] further show that over-parameterized models can make invariant learning more difficult. These results motivate several of the stress tests used in our experiments.

2.4. Model selection and diagnostic methods

Model selection is also important when evaluating OOD generalization. DomainBed [3] shows that reported performance can depend strongly on how models are selected, especially when comparing train-domain validation with OOD-based selection. In addition, DFR [10] and JTT [11] suggest that useful features may already be present in the learned representation even when the final classifier does not use them effectively. This motivates the use of linear probes in our analysis. For explainability, we use Integrated Gradients [12] as the main attribution method, while vanilla saliency is kept only as a simple baseline because of known limitations [13].

3. Problem formulation and methods

3.1. Structural causal model

Let $D \in \{0, \dots, 9\}$ denote the original MNIST digit. We first define the binary label $\tilde{Y} = \mathbf{1}[D \geq 5]$, which separates digits 0–4 from digits 5–9. To introduce label noise, the observed label is generated as $Y = \text{flip}(\tilde{Y}; 0.25)$, meaning that the binary label is flipped with probability 0.25.

The color variable C is then generated from the observed label as $C = \text{flip}(Y; p_e)$, where p_e depends on the environment e. The final image $X \in \mathbb{R}^{3\times28\times28}$ contains the original digit shape rendered in red when $C = 0$ and in green when $C = 1$.

This construction separates two sources of information. The digit shape is determined by D and remains unchanged across environments, so it acts as the stable signal. In contrast, the relationship between color C and the observed label Y depends on p_e, so color acts as the spurious signal. By changing p_e between training and test environments, we can directly study whether a model relies on the stable signal or on the easier but environment-dependent color signal.

3.2. Input

The input to the model is a colored digit image $x \in \mathbb{R}^{3\times28\times28}$. Each input contains both the original digit shape and the assigned color. The shape carries information about the underlying digit D, while the color is introduced according to the environment-specific variable C.

Because both signals are available in the same image, the model can potentially solve the task using either of them. However, only the digit shape remains reliable when the data distribution changes across environments.

3.3. Output

The task is binary classification. Given an input image x, the model estimates the probability

$$P(Y = 1 \mid X = x),$$

and produces a prediction $\hat{y} \in \{0, 1\}$.

The target label is derived from the original digit group: digits 0–4 correspond to $\tilde{Y} = 0$, while digits 5–9 correspond to $\tilde{Y} = 1$. The observed label Y is obtained after applying the 25% label noise described above. Therefore, even a classifier that perfectly recovers the digit group cannot achieve perfect accuracy with respect to Y.

3.4. Environments

We use two training environments and one OOD test environment:

$$e_1 : p_e = 0.1, \qquad e_2 : p_e = 0.2, \qquad e_{\text{te}} : p_e = 0.9.$$

The parameter p_e controls the probability that the color is flipped relative to the observed label. When p_e is small, color is strongly correlated with the label. Therefore, in the two training environments, color provides an easy predictive signal. In the OOD test environment, however, $p_e = 0.9$, so this relationship is almost reversed.

This shift creates the main difficulty of the task. A classifier that mainly uses color can perform well on the training distribution but fail when evaluated on the OOD environment. In contrast, a classifier that relies on digit shape is unaffected by the change in color correlation.

Under this construction, a color-only rule achieves approximately 0.85 ID accuracy but only around 0.10 OOD accuracy. A shape-only oracle achieves approximately 0.75 accuracy across all environments, with this upper bound caused by the 25% label noise.

3.5. Learning objectives

For each environment e, let

$$R^e(\theta) = \mathbb{E}_{(x,y)\sim P_e}\left[\ell(f_\theta(x), y)\right],$$

where ℓ denotes the binary cross-entropy loss and $R^e(\theta)$ is the risk of model f_θ in environment e. The methods considered in this report use the same environment-specific risks but optimize them using different objectives.

3.5.1. Empirical Risk Minimization (ERM)

ERM minimizes the average risk across all training environments:

$$\min_\theta \frac{1}{E}\sum_e R^e(\theta).$$

ERM does not explicitly distinguish between stable and spurious signals. Therefore, if color provides an easier predictive signal during training, the model may rely on color even when this relationship does not remain valid in the OOD environment.

3.5.2. Invariant Risk Minimization (IRMv1)

IRM aims to learn a representation for which the same classifier remains suitable across different training environments. In practice, IRMv1 introduces a gradient-based penalty:

$$\min_\Phi \frac{1}{E}\sum_e R^e(w \circ \Phi) + \lambda \sum_e \left\|\nabla_{w=1} R^e(w \cdot \Phi)\right\|^2.$$

The penalty encourages the learned representation to support a shared classifier across environments. The implementation uses the mean environment risk and the sum of the environment penalties. The penalty weight is zero before step 190 and takes its configured value from step 190 onward.

3.5.3. Variance Risk Extrapolation (V-REx)

V-REx encourages the model to achieve similar risks across training environments. Its objective is defined as

$$\min_{\theta} \operatorname{mean}_e \left(R^e\right) + \beta \cdot \operatorname{Var}_e \left(R^e\right),$$

where β controls the strength of the variance penalty. A larger value of β places more emphasis on reducing differences in risk between environments.

3.5.4. GroupDRO-lite

GroupDRO-lite focuses on the environment with the highest risk and minimizes the worst-environment loss:

$$\min_{\theta} \max_{e} R^e(\theta).$$

Unlike ERM, which optimizes the average risk, this objective gives more attention to the environment in which the model currently performs worst. GroupDRO-lite is implemented as a reference objective but is not evaluated in the reported experiments.

For IRMv1, we use the standard dummy-classifier formulation with $w = 1.0$ to compute the gradient penalty. Evaluation metrics, dataset sizes, and the model-selection protocol used throughout the experiments are described in §4.2.

3.6. Implementation and optimization

3.6.1. Pipeline

The implementation follows a simple `data`→`train`→`eval`→`viz` pipeline.

The data module, `src/cmnist/data.py`, constructs the Colored-MNIST datasets following the protocol introduced by Arjovsky et al. [2]. It also supports Colored-FashionMNIST and an optional 3×3 spurious patch used in several failure-case experiments. Each generated sample is assigned to a predefined environment according to the corresponding color-flip probability.

The training stage receives these environment-specific datasets together with a model architecture and a learning objective. The same pipeline is used across all experiments so that differences in results mainly come from the objective, model architecture, or experimental setting rather than from changes in the training procedure.

3.6.2. Models and optimization

Two backbone architectures are used. The first is a multilayer perceptron (MLP) with two hidden layers of 256 units each. The second is a small convolutional neural network (CNN), which provides a higher-capacity image-specific backbone for comparison.

The learning objectives are implemented in `losses.py` and include ERM, IRMv1, V-REx, and GroupDRO-lite. Training is handled by `train.py` using the Adam optimizer with a learning rate of 10^{-3}, a batch size of 2048, and 2000 optimization steps for the main experiments.

IRMv1 uses a penalty weight of zero before step 190 and applies its configured penalty from step 190 onward. V-REx applies its variance penalty from the first training step. The warm-up schedule therefore applies only to IRMv1.

During training, the pipeline records checkpoints according to both ID validation performance and OOD performance. These checkpoints are later used to compare the effect of different model-selection protocols.

3.6.3. Evaluation

The evaluation module, `eval.py`, measures model performance separately for each environment. The reported metrics include per-environment accuracy, worst-group accuracy, the performance gap between environments, and Expected Calibration Error (ECE).

These metrics are intended to capture different aspects of model behavior. Accuracy measures predictive performance, worst-group accuracy reflects the weakest label–color subgroup, the environment gap indicates how much performance changes under distribution shift, and ECE provides information about calibration.

Additional diagnostic outputs, including attribution maps and linear-probe results, are produced for the analysis in RQ3. Visualization utilities are separated from the training and evaluation code so that the same trained checkpoints can be reused for different diagnostic analyses.

3.6.4. Experiment budget

To manage the experimental workload, we use three execution modes.

- `fast`: 500 training steps and one random seed, used mainly to verify that the full pipeline runs correctly.
- `base`: 2000 training steps for ERM, IRMv1, and V-REx across three random seeds. This configuration is used for the main comparison in RQ1.
- `heavy`: extended experiments including penalty sweeps, backbone comparisons, failure-case variants, and transfer to Colored-FashionMNIST. This configuration contains 54 entries covering 18 settings across three seeds.

All main experiments use random seeds $\{0, 1, 2\}$. The main comparison contains nine training runs. Twelve ablation entries reuse matching main-experiment runs, leaving 42 additional training runs and 51 unique training runs overall. Deterministic cuDNN settings are enabled to reduce variation caused by non-deterministic GPU operations. The experiments were

run in the `explanable-ai` mamba environment using Python 3.11.16 and CUDA-enabled PyTorch on an NVIDIA RTX 3060 Laptop GPU with 6 GB of memory.

4. Experimental setup

4.1. Purpose of evaluation

The experimental evaluation is designed around the three research questions introduced in §1. For **RQ1 (reproduce)**, we examine how ERM behaves under distribution shift and compare it with invariant learning objectives under the same model-selection protocol. For **RQ2 (break)**, we study how the behavior of invariant learning changes when varying the penalty strength, backbone capacity, number of environments, number of spurious features, sample size, and dataset. For **RQ3 (diagnose)**, we use calibration, attribution methods, and representation probes to investigate whether the model reduces its reliance on the spurious feature.

Section 5 presents the results according to these three research questions.

4.2. Evaluation protocol

4.2.1. Data

All experiments follow the Colored-MNIST construction described in §3. The MNIST training set is divided into two disjoint training environments with approximately 30,000 images each, using $p_e = 0.1$ and $p_e = 0.2$. We reserve 20% of each environment for ID validation, leaving approximately 24,000 training images and 6,000 validation images per environment. The split is fixed across training seeds. The 10,000 MNIST test images form the OOD environment with $p_e = 0.9$.

In the small-n setting, each environment contains 6,250 images before splitting, giving 5,000 training images and 1,250 validation images. The same construction and splitting procedure is applied to FashionMNIST for the dataset-transfer experiments.

4.2.2. Metrics

We report several performance metrics. The average ID accuracy across the two training environments is

$$\text{ID-avg} = \frac{\text{Acc}_{e_1} + \text{Acc}_{e_2}}{2}.$$

OOD accuracy is the classification accuracy on the held-out test environment,

$$\text{OOD accuracy} = \text{Acc}_{e_{\text{te}}}.$$

To measure the change in performance under distribution shift, we also report the OOD gap:

$$\text{OOD gap} = \text{ID-avg} - \text{OOD accuracy}.$$

A larger gap indicates a larger reduction in accuracy when moving from the training environments to the OOD environment.

To examine performance across combinations of label and color, the test samples are divided into groups $g = (y, c) \in \{0, 1\}^2$. We report the worst-group accuracy as

$$\text{Worst-group accuracy} = \min_g \text{Acc}_g.$$

This metric indicates whether good average performance hides poor performance on one label–color subgroup. These synthetic groups do not directly measure demographic fairness. Model calibration is evaluated using Expected Calibration Error (ECE) with 15 confidence bins. For each bin, we compare mean prediction confidence with the fraction of correct predictions, then weight the absolute difference by the proportion of samples in that bin.

4.2.3. Model selection

The primary model-selection protocol is **train-domain validation**. We select the checkpoint with the highest average accuracy on the held-out ID validation environments, without using OOD test information. Training accuracy is reported separately and is not used for checkpoint selection.

For comparison, we report an **OOD oracle**, selected using OOD test accuracy, and the final checkpoint. The oracle uses test information and serves only as a diagnostic comparison. Checkpoints are evaluated every 200 steps over 2000 training steps; ties retain the first checkpoint with the highest accuracy.

Experiments use three training seeds. OOD and worst-group accuracies are reported as mean$\pm$sample standard deviation; training accuracy, gap, and ECE are reported as means in the tables. Diagnostic figures show seed 0 unless otherwise stated.

5. Results and discussion

5.1. Main comparison (RQ1)

Table 1 and Fig. 1 summarize the main comparison under train-domain validation. ERM achieves a training accuracy of 0.851 but only 0.119 ± 0.010 OOD accuracy. IRMv1 and V-REx obtain higher OOD accuracies of 0.331 ± 0.152 and 0.340 ± 0.038, respectively. Their mean results are close, although IRMv1 varies more across seeds. V-REx also achieves a higher worst-group accuracy, at 0.236 ± 0.045, compared with 0.074 ± 0.058 for IRMv1. None of the three methods achieves high OOD accuracy under train-domain selection.

Table 1: Main OOD comparison under train-domain validation (3 seeds; OOD and worst-group accuracy are mean±sample SD; other metrics are means).

Method	Train (ID avg)	Test (OOD)	OOD gap	Worst group	ECE
ERM	0.851	0.119 ± 0.010	0.732	0.005 ± 0.005	0.652
IRMv1	0.779	0.331 ± 0.152	0.448	0.074 ± 0.058	0.411
V-REx	0.820	0.340 ± 0.038	0.480	0.236 ± 0.045	0.340

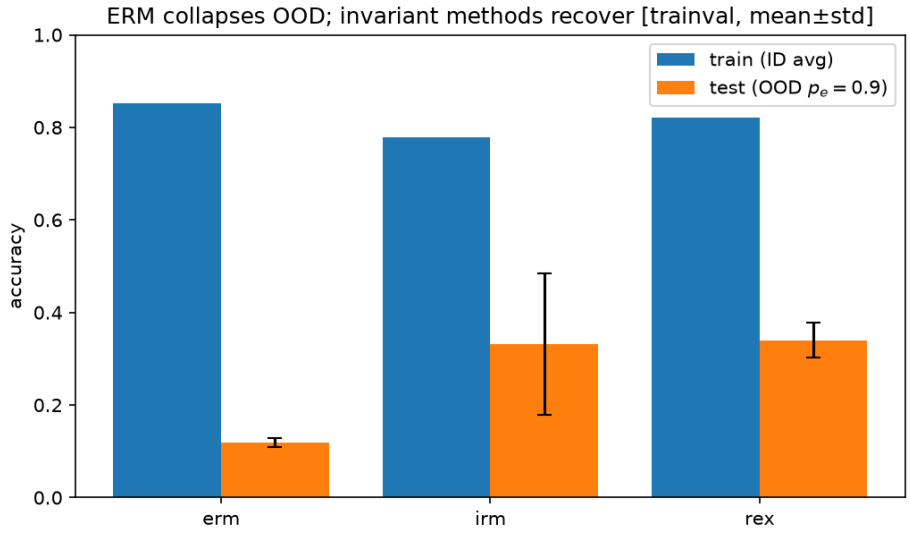


Figure 1: Training and OOD test accuracy under train-domain validation. Error bars on the OOD series show sample standard deviation over three training seeds. Checkpoints are selected using held-out ID validation.

5.1.1. Effect of model selection

The choice of model-selection protocol also affects the reported OOD performance. Table 2 compares train-domain validation, OOD-oracle selection, and the final checkpoint, following the distinction discussed in DomainBed [3].

All three methods obtain higher OOD accuracy under oracle selection. ERM increases from 0.119 ± 0.010 to 0.338 ± 0.020, while V-REx increases from 0.340 ± 0.038 to 0.572 ± 0.028. The largest difference occurs for IRMv1, which increases from 0.331 ± 0.152 to 0.759 ± 0.075. Its final checkpoint obtains 0.650 ± 0.108.

Table 2: OOD accuracy under different model-selection protocols (mean±sample SD over 3 seeds).

Method	Train-domain validation	OOD oracle	Last checkpoint
ERM	0.119 ± 0.010	0.338 ± 0.020	0.280 ± 0.017
IRMv1	0.331 ± 0.152	0.759 ± 0.075	0.650 ± 0.108
V-REx	0.340 ± 0.038	0.572 ± 0.028	0.360 ± 0.065

IRMv1 therefore reaches checkpoints with substantially better OOD performance than those selected by ID validation. However, selecting these checkpoints using OOD test accuracy requires information unavailable under the primary evaluation protocol.

5.2. Penalty stability (RQ2a)

Table 3 summarizes the penalty-strength sweep over three random seeds. OOD and worst-group accuracies are reported as arithmetic mean and sample standard deviation ($n-1$ denominator); the other metrics are means.

Table 3: Penalty-strength sweep under train-domain validation (3 seeds; OOD and worst-group accuracy are mean±sample SD; other metrics are means).

Method	Penalty	Train	Test OOD	Gap	Worst	ECE
IRMv1	$\lambda = 100$	0.829	0.284 ± 0.067	0.545	0.103 ± 0.083	0.404
IRMv1	$\lambda = 1000$	0.778	0.337 ± 0.153	0.441	0.081 ± 0.064	0.405
IRMv1	$\lambda = 10000$	0.779	0.331 ± 0.152	0.448	0.074 ± 0.058	0.411
IRMv1	$\lambda = 30000$	0.780	0.331 ± 0.151	0.449	0.074 ± 0.058	0.411
V-REx	$\beta = 100$	0.872	0.221 ± 0.020	0.651	0.118 ± 0.021	0.559
V-REx	$\beta = 1000$	0.820	0.340 ± 0.038	0.480	0.236 ± 0.045	0.340
V-REx	$\beta = 10000$	0.797	0.417 ± 0.045	0.379	0.297 ± 0.099	0.146

Table 4 reports the individual OOD and worst-group accuracies underlying these averages. Each row corresponds to one method and penalty strength; its three seed columns are separate training runs.

Table 4: OOD and worst-group accuracy across seeds and penalty strengths.

Method	Penalty	Seed 0 OOD / Worst	Seed 1 OOD / Worst	Seed 2 OOD / Worst
IRMv1	$\lambda = 100$	0.3048 / 0.1621	0.2083 / 0.0083	0.3379 / 0.1378
IRMv1	$\lambda = 1000$	0.2817 / 0.1051	0.2188 / 0.0078	0.5093 / 0.1291
IRMv1	$\lambda = 10000$	0.2765 / 0.0968	0.2137 / 0.0080	0.5035 / 0.1165
IRMv1	$\lambda = 30000$	0.2796 / 0.1011	0.2123 / 0.0078	0.5017 / 0.1131
V-REx	$\beta = 100$	0.1985 / 0.0945	0.2332 / 0.1222	0.2313 / 0.1360
V-REx	$\beta = 1000$	0.2962 / 0.1841	0.3668 / 0.2580	0.3577 / 0.2651
V-REx	$\beta = 10000$	0.4175 / 0.3374	0.4627 / 0.3691	0.3722 / 0.1837

For IRMv1, increasing λ from 10^3 to 3×10^4 changes OOD accuracy relatively little within each seed. Over this range, seed 0 obtains approximately 0.276–0.282, seed 1 obtains 0.212–0.219, and seed 2 obtains 0.502–0.509. Differences between seeds are therefore larger than the differences between these penalty strengths.

The $\lambda = 10^2$ setting gives a mean OOD accuracy of 0.284 ± 0.067. Higher penalties increase the mean slightly but also show greater variation across seeds. The sweep does not show a consistent improvement from increasing the IRMv1 penalty.

For V-REx, mean OOD accuracy increases from 0.221 ± 0.020 at $\beta = 10^2$ to 0.340 ± 0.038 at $\beta = 10^3$ and 0.417 ± 0.045 at $\beta = 10^4$. Worst-group accuracy also increases, reaching 0.297 ± 0.099 at the highest tested penalty. A stronger V-REx penalty improves average performance in this sweep, but it does not reduce variation across seeds. The highest penalty gives the best mean OOD accuracy among the tested V-REx settings.

5.3. Backbones, failure cases, and transfer (RQ2b)

These experiments examine how changes to model architecture, training environments, and data affect the behavior observed in the main Colored-MNIST experiment. Table 5 first examines backbone architecture and the number of training environments.

Table 5: Backbone and environment ablations under train-domain validation (3 seeds; OOD and worst-group accuracy are mean±sample SD; other metrics are means).

Method	Configuration	Train	Test OOD	Gap	Worst	ECE
ERM	MLP	0.851	0.119 ± 0.010	0.732	0.005 ± 0.005	0.652
ERM	CNN	0.849	0.108 ± 0.004	0.740	0.003 ± 0.004	0.660
IRMv1	MLP	0.779	0.331 ± 0.152	0.448	0.074 ± 0.058	0.411
IRMv1	CNN	0.593	0.599 ± 0.009	-0.007	0.385 ± 0.072	0.096
IRMv1	Single training environment	0.693	0.529 ± 0.233	0.164	0.120 ± 0.153	0.363

5.3.1. Backbone architecture

With IRMv1, the CNN obtains an OOD accuracy of 0.599 ± 0.009, compared with 0.331 ± 0.152 for the MLP (Table 5). Worst-group accuracy also increases, from 0.074 ± 0.058 to 0.385 ± 0.072. The CNN therefore performs better on both measures and shows less variation in OOD accuracy. These results show that the backbone affects the outcome of IRMv1 in this setting; they do not support the claim that the tested CNN causes invariant learning to fail.

5.3.2. Single training environment

IRMv1 trained with one environment obtains 0.529 ± 0.233 OOD accuracy, compared with 0.331 ± 0.152 with two environments. The single-environment result has a higher mean but also greater variation across seeds. It therefore does not show that using one environment necessarily reduces average OOD accuracy.

This comparison also changes the total amount of training data: the single-environment setting uses approximately 24,000 training images, compared with 48,000 across two environments. The experiment does not isolate the effect of environment count from sample size.

Table 6 examines additional spurious features, reduced sample size, and transfer to Colored-FashionMNIST.

Table 6: Spurious-feature, sample-size, and dataset-transfer ablations under train-domain validation (3 seeds; OOD and worst-group accuracy are mean±sample SD; other metrics are means).

Method	Configuration	Train	Test OOD	Gap	Worst	ECE
ERM	Additional 3×3 patch	0.930	0.138 ± 0.002	0.792	0.078 ± 0.001	0.786
IRMv1	Additional 3×3 patch	0.908	0.114 ± 0.002	0.794	0.095 ± 0.001	0.791
ERM	$n = 6{,}250$ per environment	0.959	0.255 ± 0.027	0.704	0.164 ± 0.023	0.612
IRMv1	$n = 6{,}250$ per environment	0.810	0.333 ± 0.102	0.478	0.051 ± 0.076	0.462
ERM	Colored-FashionMNIST	0.849	0.103 ± 0.001	0.747	0.000 ± 0.001	0.651
IRMv1	Colored-FashionMNIST	0.710	0.394 ± 0.266	0.315	0.092 ± 0.134	0.372

5.3.3. Additional spurious feature

We introduce a second spurious feature in the form of a 3×3 patch (Table 6). Adding the patch reduces IRMv1 OOD accuracy to 0.114 ± 0.002. ERM obtains 0.138 ± 0.002 in the same setting. Both methods perform poorly despite high training accuracy.

The additional spurious feature makes the tested setting more difficult for IRMv1. This experiment is relevant to the limitations discussed by Rosenfeld et al. [9], although it does not directly verify their theoretical conditions.

5.3.4. Limited training data

With 6,250 images per environment before splitting, IRMv1 obtains 0.333 ± 0.102 OOD accuracy, close to 0.331 ± 0.152 in the standard setting. Its OOD standard deviation decreases, while mean worst-group accuracy falls from 0.074 to 0.051.

ERM OOD accuracy increases from 0.119 ± 0.010 to 0.255 ± 0.027. Reducing the sample size therefore does not produce a uniform decline in performance or stability across the tested methods.

5.3.5. Transfer to Colored-FashionMNIST

On Colored-FashionMNIST, IRMv1 obtains 0.394 ± 0.266 OOD accuracy, compared with 0.103 ± 0.001 for ERM. IRMv1 has a higher mean, but its large standard deviation shows considerable variation across the three seeds.

Overall, the ablations show that the observed results depend on architecture, penalty strength, and the available signals. The additional patch substantially reduces IRMv1 performance, while the CNN improves it. Fewer environments and fewer samples do not consistently reduce average OOD accuracy in these experiments.

5.4. Diagnostics (RQ3)

The previous experiments evaluate the models mainly through predictive performance. To better understand how the models use the available information, we also examine their calibration, input attributions, recoloring sensitivity, and learned representations.

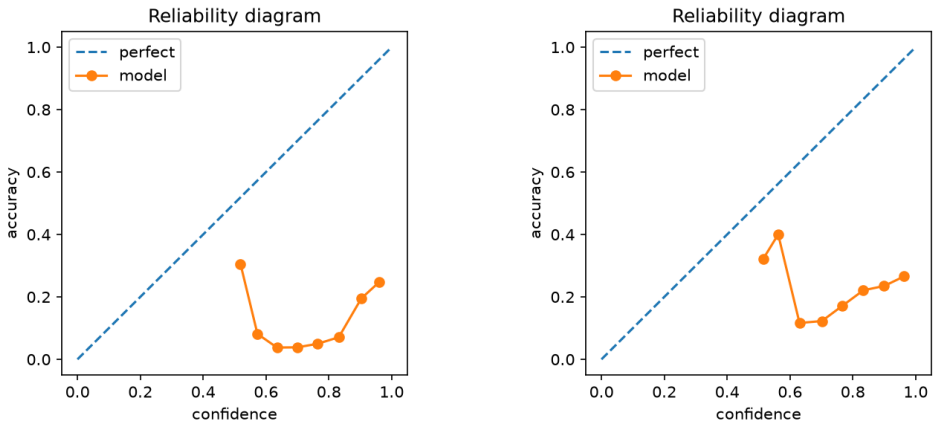
5.4.1. Calibration

Across three seeds, mean OOD ECE is 0.652 for ERM, 0.411 for IRMv1, and 0.340 for V-REx. The invariant objectives therefore reduce ECE relative to ERM, although calibration errors remain substantial.

Figure 2 shows seed 0, where ERM and IRMv1 obtain ECE values of 0.664 and 0.376. These values describe the plotted checkpoints and differ from the three-seed averages in the main table.

5.4.2. Attribution analysis

Integrated Gradients highlights spatial contributions to the predictions for the same four test images (Fig. 3). Since color is carried by the digit strokes, the channel-summed maps



(a) ERM (seed 0, ECE 0.664). (b) IRMv1 (seed 0, ECE 0.376).

Figure 2: Reliability diagrams on 10,000 OOD test images for seed 0, selected by held-out ID validation. The diagonal represents perfect calibration. Confidence is compared with the fraction of correct predictions in each bin.

cannot separate dependence on shape from dependence on color. The maps are normalized separately, so their intensities should not be compared as absolute attribution magnitudes.

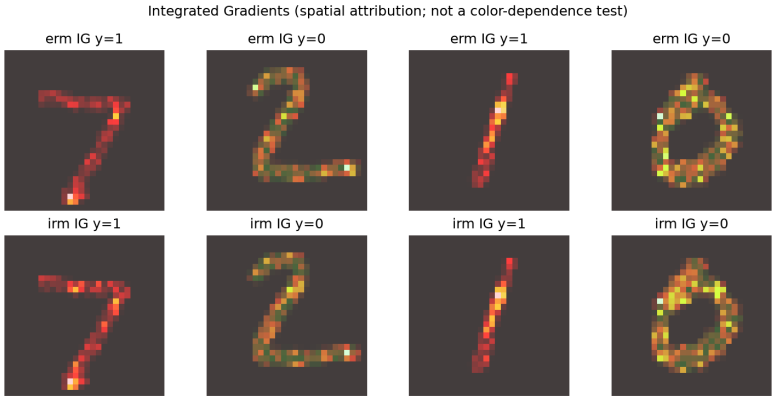


Figure 3: Integrated Gradients for ERM (top row) and IRMv1 (bottom row), using seed-0 held-out-ID-selected checkpoints and a zero baseline with 32 integration steps. Each channel-summed map is scaled separately and overlaid on its input image; the maps do not separate shape dependence from color dependence.

For reference, we also report vanilla saliency maps in Fig. 4. However, vanilla saliency is used only as a secondary baseline because gradient-based saliency methods are known to fail several sanity checks [13].

5.4.3. Recoloring sensitivity

To examine sensitivity to color more directly, we swap the red and green channels while preserving the digit shape. The mean prediction-flip rate is 0.972 ± 0.016 for ERM, 0.598 ± 0.321 for IRMv1, and 0.630 ± 0.058 for V-REx. IRMv1 and V-REx change their predictions less often on average, but both remain sensitive to recoloring.

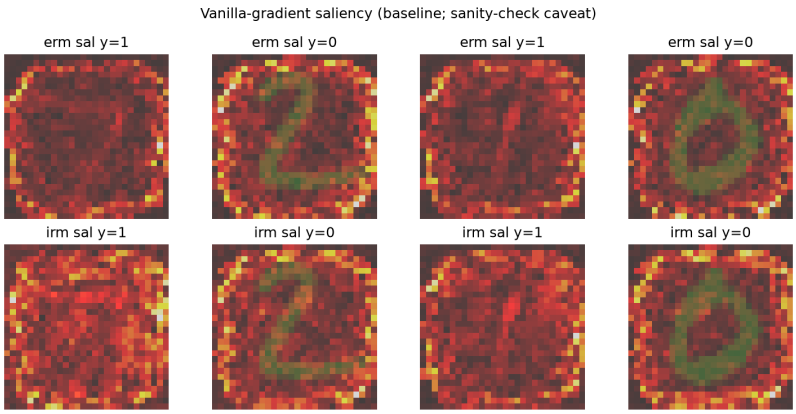


Figure 4: Vanilla-gradient saliency for ERM (top row) and IRMv1 (bottom row), using the same seed-0 checkpoints and images as Fig. 3. Each map is scaled separately and overlaid on its input image. The maps show spatial sensitivity and do not separate shape dependence from color dependence.

5.4.4. Representation probe

A linear color probe achieves held-out accuracy of 1.00 for both ERM and IRMv1 across the three seeds. Color therefore remains decodable from both representations. Together with the recoloring results, this shows that retaining color information and responding to color changes are different properties. These diagnostics do not establish which mechanism in the final classification layer produces the difference.

6. Conclusion

This report compared ERM, IRMv1, and V-REx on Colored-MNIST using three random seeds and a common evaluation pipeline.

For **RQ1**, ERM achieved high training accuracy but only 0.119 ± 0.010 OOD accuracy. IRMv1 and V-REx improved the mean OOD result, although their performance remained limited under train-domain selection. IRMv1 was particularly sensitive to checkpoint selection, increasing from 0.331 ± 0.152 under ID validation to 0.759 ± 0.075 under OOD-oracle selection.

For **RQ2**, larger IRMv1 penalties did not consistently improve performance. V-REx obtained its highest mean OOD accuracy at $\beta = 10^4$, but this setting did not have the lowest variation. The CNN improved IRMv1 performance, while an additional spurious patch reduced it substantially. The single-environment and small-sample experiments did not show a consistent decrease in average OOD accuracy.

For **RQ3**, calibration, recoloring tests, and representation probes provided complementary information. Color remained decodable from both ERM and IRMv1 representations. The attribution maps showed spatial contributions but could not establish a preference for shape over color.

These conclusions are limited to the tested synthetic shifts, configurations, and three training seeds. They do not establish a general ranking of invariant learning methods. Real-world datasets such as Waterbirds or CelebA, larger models, and broader hyperparameter searches remain directions for future work.

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A. Reproducibility

The source code, experiment configurations, and scripts used in this report are available at:

`https://github.com/namng194/spurious-correlations-invariant-learning`

A.1. Environment setup

The experiments were run in the `explanable-ai` mamba environment with Python 3.11.16 and PyTorch 2.14.1+cu130. CUDA was enabled on an NVIDIA RTX 3060 Laptop GPU with 6 GB of memory. The following commands use Windows CMD and assume that the project dependencies are installed in this environment:

```
mamba activate explanable-ai
python -m pytest tests/ -x -q
set RESULTS=outputs/corrected_20261016
```

A.2. Main experiments

The main and ablation experiments use the base and heavy configurations, respectively. Matching main runs are reused in the ablation study:

```
python scripts/run_sweep.py ^
  --config configs/base.yaml ^
  --out %RESULTS%/sweep --resume
python scripts/ablate.py ^
  --config configs/heavy.yaml ^
  --out %RESULTS%/ablate --resume ^
  --reuse-sweep %RESULTS%/sweep
```

A.3. Figures and diagnostics

The report uses the existing figures in `outputs/corrected_20261016/figs/`, copied into `report/figs/`. The following commands describe how they were generated; they need not be rerun when only the report text is changed:

```
python scripts/make_plots.py ^
  --in %RESULTS%/sweep/results.json ^
  --out %RESULTS%/figs
python scripts/make_diagnostics.py ^
  --config configs/base.yaml --seed 0 ^
  --checkpoint-root %RESULTS%/sweep ^
  --out %RESULTS%/figs
```

Diagnostics reuse the held-out-ID-selected checkpoints rather than training separate models. The same diagnostic command is run with `-seed 1` and `-seed 2`, writing to `diagnostics/seed1/` and `diagnostics/seed2/` under the results directory. These three diagnostic outputs provide the averages reported in the text; the included figures use seed 0.

A.4. Saved results

The main result files are stored under `outputs/corrected_20261016/`:

- `sweep/results.json`: main results for all checkpoint-selection rules;
- `ablate/ablate.json`: the 54 ablation entries;
- `figs/diagnostics.json`: seed-0 diagnostics;
- `diagnostics/seed1/diagnostics.json` and `diagnostics/seed2/diagnostics.json`: the remaining seeds;
- `diagnostics_summary.json`: three-seed ECE and recoloring summaries.

Individual training directories also contain configuration, split indices, training history, selected steps, and saved checkpoints. These artifacts provide the values used in Section 5.